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3. Problem setting. A scalar advection-diffusion problem without reaction.

$$(3.1) \quad u_t + \nabla \cdot (F(u) - D(u)\nabla u) = 0, \mathbf{x} \in \mathbb{R}^d$$

The variable $D(u)$ will be treated with Kirchhoff transformation. The diffusion coefficient D is treated as a constant here. This will not affect the following discussion on numerical scheme.

$$(3.2) \quad u_t + \nabla \cdot F(u) - D \Delta u = 0, \mathbf{x} \in \mathbb{R}^d$$

3.1. Finite volume scheme. Replace scalar u with cell averaged value

$$(3.3) \quad \bar{u} = \frac{1}{|E|} \int_E u(\mathbf{x}, t) d\mathbf{x}$$

and Lax-Friedrichs numerical flux

$$(3.4) \quad f(u) = \frac{1}{2} [F(u^+) + F(u^-) \cdot \mathbf{v}_E - \alpha(u^+ - u^-)]$$

The finite volume scheme of advection-diffusion problem is:

$$(3.5) \quad \bar{u}_t + \frac{1}{|E|} \int_{\partial E} (\hat{f}(\bar{u}) - \nabla \bar{u} \cdot \mathbf{v}_E) d\sigma(\mathbf{x}) = 0$$

In 2D quadrilateral logically rectangular mesh, we apply gaussian quadrature every edge as the approximation of integral of flux. I use Multi-level WENO reconstruction scheme to reconstruct the point value in advection numerical flux.

$$(3.6) \quad \hat{f}(u) = \frac{1}{2} [F(\mathcal{R}^+(\{\bar{u}\}_+)) + F(\mathcal{R}^-(\{\bar{u}\}_-)) \cdot \mathbf{v}_E - \alpha(\mathcal{R}^+(\{\bar{u}\}_+) - \mathcal{R}^-(\{\bar{u}\}_-))]$$

Let $\{\bar{u}\}_+ = \{\bar{u}_i \mid i \in \text{WENO stencil } I_+\}$. Diffusion flux is different from its advection counterpart. I reconstruct derivative across the edge by sampling along the normal direction.

4. Computation of Jacobian. Implicit method contains solving a non linear system and a linear subproblem. I write the general non linear system as:

$$(4.1) \quad \bar{u}_t + \mathcal{F}(\bar{u}) - \mathcal{G}(\bar{u}) = 0$$

where $\mathcal{F}(\bar{u})$ represents advection part, and $\mathcal{G}(\bar{u})$ represents diffusion part. The semi discretized form is

$$(4.2) \quad \bar{u}^{n+1} - \bar{u}^n + \Delta t [\mathcal{F}(\bar{u}^{n+1}) - \mathcal{G}(\bar{u}^{n+1})] = 0$$

There are many well-researched numerical methods that can be used to solve such non linear system.

4.1. ML-WENO reconstruction and derivatives. I use ML-WENO reconstruction scheme to reconstruct point values from cell-averaged value... I use two stage non linear weighting. The first stage is calculated as

$$(4.3) \quad \check{\omega}_l = \frac{\omega_l}{(\sigma_l + \varepsilon h_0^2) \eta_l} \quad \eta_l = \begin{cases} 1, & \text{if } r = 1 \\ 3, & \text{if } r = 2 \\ 4, & \text{if } r \geq 3 \end{cases}$$

$$(4.4) \quad \bar{\omega}_l = \frac{\ddot{\omega}_l}{\sum_k \ddot{\omega}_k}$$

The the second stage is calculated based on first stage:

$$(4.5) \quad \hat{\omega}_l = \frac{\bar{\omega}_l}{(\sigma_l + \varepsilon h_0^2)^{sr_l}}$$

$$(4.6) \quad \tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_k \hat{\omega}_k}$$

The reconstruction is expressed as the weighted linear combination of stencil polynomials.

$$(4.7) \quad \mathcal{R}(\mathbf{x}) = \sum_{l \in L} P_l(\mathbf{x}) \tilde{\omega}_l(\bar{u})$$

Suppose the reconstruction happens in a given stencil I . The cell averaged values $\bar{u}$ in this stencil is represented as $\{\bar{u}_s\}_{s \in S}$. The derivative of this reconstruction with respect to $\bar{u}_s$ is:

$$(4.8) \quad \frac{\partial \mathcal{R}}{\partial \bar{u}_s} = \sum_{l \in L} \left(\frac{\partial P_l}{\partial \bar{u}_s} \tilde{\omega}_l + P_l \frac{\partial \tilde{\omega}_l}{\partial \bar{u}_s} \right)$$

It consists of two parts, derivative of stencil polynomials $\frac{\partial P_l}{\partial \bar{u}_s}$ and derivative of non linear weights $\frac{\partial \tilde{\omega}_l}{\partial \bar{u}_s}$. The stencil polynomial is a combination of single polynomials satisfying

$$(4.9) \quad \frac{1}{|E_k|} \int_{E_k} p_s(\mathbf{x}) d\mathbf{x} = \begin{cases} 1 & s = k \\ 0 & s \neq k \end{cases}$$

I use tensorproduct polynomials for logically rectangular mesh. The tensorproduct polynomial is written in the following form. Coefficients c_α are determined by solving a linear system satisfying (??).

$$(4.10) \quad p_s = \sum_{i < I} a_i^s \left(\frac{x - x_S}{h_S} \right)^i \sum_{j < J} b_j^s \left(\frac{y - y_S}{h_S} \right)^j = \sum_{|\alpha| \leq (I-1) \times (J-1)} \tilde{c}_\alpha^s \left(\frac{\mathbf{x} - \mathbf{x}_S}{h_S} \right)^\alpha$$

The stencil polynomial is expressed in terms of a combination of all these polynomials.

$$(4.11) \quad P_l(\mathbf{x}) = \sum_{s \in S} p_s^l(\mathbf{x}) \bar{u}_s$$

Combining cell averaged values $\bar{u}_s$ with coefficients

$$(4.12) \quad c_\alpha = \sum_{s \in S} \tilde{c}_\alpha^s \bar{u}_s$$

The derivative of polynomial is straightforward. The derivative of non linear weight is a little more complicated. Equation ?? can be rewritten as:

$$(4.13) \quad \frac{\partial \mathcal{R}}{\partial \bar{u}_s} = \sum_{l \in L} p_s^l(\mathbf{x}) \tilde{\omega}_l + \sum_{l \in L} P_l \frac{\partial \tilde{\omega}_l}{\partial \bar{u}_s}$$

The derivatives of non linear weights are:

$$\begin{aligned}
\frac{\partial \tilde{\omega}_l}{\partial \bar{u}_s} &= \frac{\hat{\omega}_{l,s}}{\sum_k \hat{\omega}_k} - \frac{\hat{\omega}_l \sum_k \hat{\omega}_{k,s}}{(\sum_k \hat{\omega}_k)^2} \\
\frac{\partial \hat{\omega}_l}{\partial \bar{u}_s} &= \frac{\bar{\omega}_{l,s}}{(\sigma_l + \varepsilon h_0^2)^{sr_l}} - sr_l \frac{\bar{\omega}_l \sigma_{l,s}}{(\sigma_l + \varepsilon h_0^2)^{sr_l+1}} \\
\frac{\partial \bar{\omega}_l}{\partial \bar{u}_s} &= \frac{\hat{\omega}_{l,s}}{\sum_k \hat{\omega}_k} - \frac{\hat{\omega}_l \sum_k \check{\omega}_{k,s}}{(\sum_k \check{\omega}_k)^2} \\
\frac{\partial \check{\omega}_l}{\partial \bar{u}_s} &= -\eta_l \sigma_{l,s} \frac{\omega_l}{(\sigma_l + \varepsilon h_0^2)^{\eta_l+1}}
\end{aligned}
\tag{4.14}$$

We calculate the derivatives of non linear weights with chain rule. We also need the derivative of smoothness indicator. The Jiang-Shu smoothness indicator is:

$$\sigma_{JS} = \sum_{0 < |\beta| < r} \frac{h_0^{2|\beta|}}{|E_0|} \int_{E_0} \left(\sum_{|\alpha| < r} c_\alpha \mathcal{D}^\beta \left(\frac{\mathbf{x} - \mathbf{x}_S}{h_S} \right)^\alpha \right)^2 d\mathbf{x}
\tag{4.15}$$

where order $r = (I - 1) \times (J - 1)$, a constant determined by the size of stencil. We present an approximation of the Jiang-Shu smoothness indicator (cite) by omitting all cross product terms.

$$\begin{aligned}
\tilde{\sigma}_P &= \sum_{0 < |\alpha| < r} \sum_{0 < \beta \leq \alpha} \left(\frac{h_0}{h_S} \right)^{2|\beta|} c_\alpha^2 \left(\frac{\alpha!}{(\alpha - \beta)!} \right)^2 \frac{1}{|E_0|} \int_{E_0} \left(\frac{\mathbf{x} - \mathbf{x}_S}{h_S} \right)^{2(\alpha - \beta)} d\mathbf{x} \\
&= \sum_{0 < |\alpha| < r} \sum_{0 < \beta \leq \alpha} \mathcal{C}(\alpha, \beta) c_\alpha^2
\end{aligned}
\tag{4.16}$$

the derivative of $\tilde{\sigma}_P$ can be written as:

$$\begin{aligned}
\frac{\partial \tilde{\sigma}_P}{\partial \bar{u}_s} &= \sum_{0 < |\alpha| < r} \sum_{0 < \beta \leq \alpha} \mathcal{C}(\alpha, \beta) 2c_\alpha \frac{\partial c_\alpha}{\partial \bar{u}_s} \\
\frac{\partial c_\alpha}{\partial \bar{u}_s} &= \tilde{c}_\alpha^s
\end{aligned}
\tag{4.17}$$

4.2. Derivative of flux. Implicit method requires calculation of derivatives of both advection and diffusion flux.

4.2.1. Derivative of advection flux. Differentiate equation ?? is straight forward.

$$\begin{aligned}
\frac{\partial \hat{f}(u)}{\partial \bar{u}_s} &= \frac{1}{2} \left[\left(\frac{\partial \mathcal{R}^+}{\partial \bar{u}_s} + \frac{\partial \mathcal{R}^-}{\partial \bar{u}_s} \right) \frac{\partial F}{\partial \bar{u}_s} \cdot \mathbf{v}_E - \alpha \left(\frac{\partial \mathcal{R}^+}{\partial \bar{u}_s} - \frac{\partial \mathcal{R}^-}{\partial \bar{u}_s} \right) \right] \\
&= \frac{1}{2} \left[\frac{\partial \mathcal{R}^+}{\partial \bar{u}_s} \left(\frac{\partial F}{\partial \bar{u}_s} \cdot \mathbf{v}_E - \alpha \right) + \frac{\partial \mathcal{R}^-}{\partial \bar{u}_s} \left(\frac{\partial F}{\partial \bar{u}_s} \cdot \mathbf{v}_E + \alpha \right) \right]
\end{aligned}
\tag{4.18}$$

4.2.2. Derivative of diffusion flux.

5. Implicit time stepping.

5.1. Backward Euler.

5.2. Implicit SSP time stepping.

6. Parallelism. Parallel solver is inevitable in large scale computation. We talk about mesh partitioning strategy across different processors in the following sections.

6.1. Logically rectangular mesh and stencils. Cells in logically rectangular mesh maintains logical order. Logically rectangular mesh shows more flexibility in some computation scenerios than cartesian mesh. Figure ?? shows a full view of a logically rectangular (quadrilateral) mesh. Counting all stencils without repeating is convenient in the logically rectan-

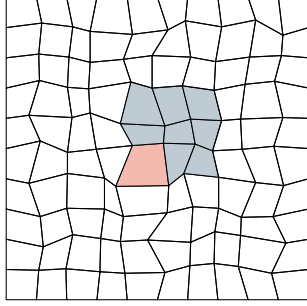


Fig. 6.1. Logically rectangular mesh displayed in full. The index of a stencil (colored cells) is the same as the index of the red cell.

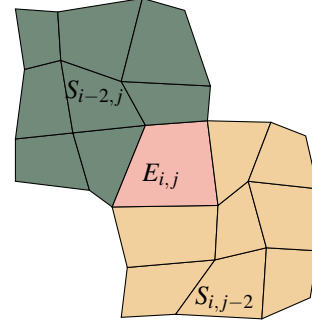


Fig. 6.2. Two sample stencils used by cell $E_{i,j}$. The index of these two stencils are $S_{i-2,j}$ and $S_{i,j-2}$. Consistent with the index of bottom left cell.

gular mesh setting. We use the index of bottom left cell as the index of the corresponding stencil. A $M \times N$ mesh will then contain $(M - I + 1) \times (N - J + 1)$ stencils of size $I \times J$.

6.2. Mesh partition. I examine the following partition of mesh without lossing generality. The dashed line indicates ghost cells distributed by MPI communication. The size of ghost layer is determined by the largest stencil size required for cells in the ghost layer. Increasing the order of reconstruction (increasing the size of corresponding stencil) will thus increase the communication cost. However, the distribution of mesh only occur once in the beginning of the computation. Mesh will not change during the process of compuation. Figure ?? shows a stencil covers ghost layer.

6.3. Sparsity pattern of Jacobian matrix.

6.4. Optimal preconditioner.

- 1. Incomplete LU decomposition
- 2. Additive schwarz

6.5. Scalability.

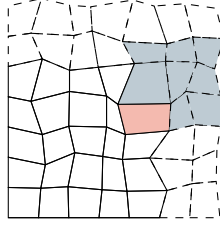


Fig. 6.3. Illustration of mesh partition. Ghost cells are drawn with dashed lines. The size of ghost layer is determined by the number of cells need in the time stepping computation.

7. Magma migration. For our very first simulation, we would like to simulate the production of magma in a simplified controlled scenerio. I was inspired by the setup described previously in[5]. The simulation will be setup in a chunk of rock ascending with constant upwelling velocity. A linear adiabatic temperature profile is applied here. A linear intial rock composition is also applied here.

7.1. Mathematical model. We are going to show our coupled darcy-stokes semi-degenerate model can solve this problem well. In[5], matrix creeping (Stokes equation) did not directly involved in the simulation due to the fact that boundary condition restricts a constant background upwelling ascending velocity. The solid matrix pressure is therefore strictly lithostatic pressure relating only to depth. In[5], only energy transports while concentration does not participate in transport phenomenon.

$$\begin{aligned}
 \check{\mathbf{v}}_r + \phi_f^{1+\Theta} \nabla (\phi_f^{-1/2} \check{q}_f) &= 0 \\
 \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \check{\mathbf{v}}_r) + \frac{1}{1-\phi_f} (\check{q}_f - \phi_f^{1/2} \check{q}) &= 0 \\
 \nabla \check{q} - \nabla \cdot \hat{\sigma}(\check{\mathbf{v}}_s) &= -(1-\phi_f) \hat{\mathbf{g}} \\
 \nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f} (\check{q}_f - \phi_f^{1/2} \check{q}) &= 0
 \end{aligned}
 \tag{7.1}$$

For simplification, we won't include diffusion into our transport model. We drop the diffusion term here for the fact that diffusion coefficient is relatively small compared to convective phenomenon. On the other hand, eutectic melting has a constant concentration such that gradient of concentration is of zero value. Hence the transport of component 2 (opx) can be expressed in terms of mixture theory as:

$$\begin{aligned}
 \bar{C}_t + \nabla \cdot (\bar{C} \mathbf{v}) &= 0 \\
 \bar{C}_t + \nabla \cdot (\mathbf{v}_e^* \bar{C}) &= 0
 \end{aligned}
 \tag{7.2}$$

where $\mathbf{v}_e^*$ denotes effective veclocity $\mathbf{v}_e^* = \kappa \mathbf{v}_f + (1-\kappa) \mathbf{v}_s$ and $\kappa = \frac{\phi_f C_f}{\bar{C}}$ where C_f and $\bar{C}$ using values from previous time step. Enthalpy is transported in a similar manner.

$$\bar{\mathcal{H}}_t + \nabla \cdot (\bar{\mathcal{H}} \mathbf{v} - \bar{K} \nabla \mathcal{T}) = 0
 \tag{7.3}$$

7.1.1. Boundary conditions. In order to solve the proposed darcy-stokes coupled equation, we need appropriate boundary condions defined around the computational domain for stokes and darcy subproblems. The behavior of solid matrix is governed by Stokes equation. Bernardi-raugel style element is used for this Stokes system, hence we need to provide proper

boundary conditions for two dofs per nodal and one additional supplemental flux bubble function for each edge dof. The corresponding boundary condition are shown as follows.

- Prescribe essential boundary condition of constant upwelling ascending velocity at supplying depth $z = z_0$ with $\mathbf{v}_s = \mathbf{W}$.
- Prescribe essential boundary condition to normal velocity and supplemental flux with the value of zero on two sides of the physical domain.
- Prescribe zero traction (free dof) boundary condition (natural boundary condition) to tangential velocity on two sides of the physical domain.
- Prescribe natural boundary condition to normal velocity and supplemental flux on the top side of the physical domain representing a free outlet scenerio.
- Prescribe essential boundary condition to tangential velocity with value zero on the top side of the physical domain.

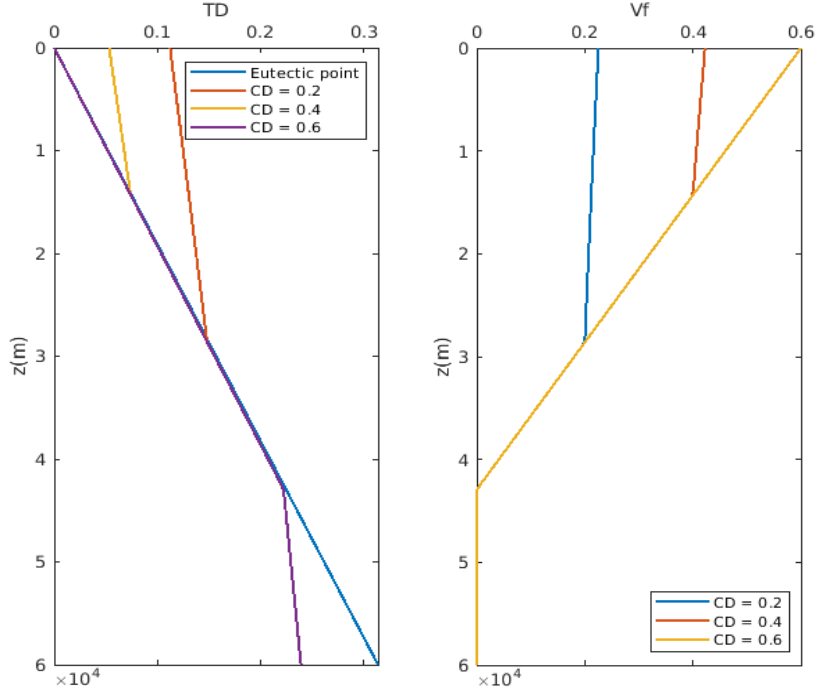
The behavior do molten matrix is governed by Darcy equation. We used BDM sytle element for this Darcy system, hence we just need to provide proper boundary condition for each edges. The corresponding boundary condition are shown as follows:

- Prescribe zero values on two sides of the physical domain.
- Prescribe zero values on the bottom of the physical domain representing no melting enters the computational domain form bottom.
- Prescribe free (natural) boundary condition on the top of the physical domain representing a free outlet of the melt.

7.1.2. Initial conditions. At the same time, we need an initial conditions for our transport equations. In[5], temperature and composition are distributed linearly along against the depth. Unlike [5], we implement eutectic melting model into our simulation. In our situation, temperature and total composition won't be deterministic. Here I make some modification on the setup. Initially, the entire domain was burried under the partial melting region, where temperature was under the sub-solidus region. Melting starts when the entire chunk ascending adiabatically. As pressure continuously lowering, the eutectic point is lowering at the same time. Hence, melting flux supplies from top.

In eutectic melting situation, we cannot impose simple linear temperature directly due to the existence of the eutectic point. Therefore, an appropriate initial condition should be assigned with careful selection.

Linear Enthalpy distribution (Near constant).



7.1.3. Melting depth. As the magma chunk ascending adiabatically, we can define a depth z_m where melting starts. From the plots above, we can conclude that in our model

- Change of dimensionless composition can only affect the separation of eutecton and super-eutecton region.
- Change of dimensionless composition will not affect where magma leave eutecton melting region.

A fixed melting depth can therefore be calculated despite dimensionless composition distribution. The melting happens when current temperature reaches the eutectic melting temperature. In the situation of linear dimensionless enthalpy distribution $f(HD) = HD_0 + \eta z$. Assuming lithostatic pressure distribution, the eutectic point at given depth z is: $\mathcal{T}_e = 0 + \gamma^* \rho_s g z$. The melting depth is therefore calculated as: $z = \frac{HD_0}{\rho_s g - \eta}$. In this case, the smaller initial dimensionless enthalpy HD_0 the shorter melting depth.

7.2. Interpolation of stokes pressure. We are using piece-wise constant L2 approximation space for stokes and darcy pressure. In order to evaluate point-wise pressure. We are going to present an interpolation using generated piece-wise constant pressure.

7.3. Approximation of physical values on the cell edges. In the process of simulation, plenty of physical values are defined on the gauss points of cell edges. Take κ for example, where $\kappa = \frac{\phi_f c_f}{\bar{c}}$, two phase parameters ϕ_f and c_f are computed with phase package using reconstructed values on both side of the cell edges. The value of $\bar{c}$ are directly obtained by WENO reconstruction. Therefore, two different values κ_{In} and κ_{Out} will be obtained during this computation process, which will be used in the calculation of the numerical flux.

7.4. Nondimensionalization and retrieve of original physical variables. We converting scaled and nondimensionalized computational variables back to original physical variables.

Note, the characteristic length defined similarly to the definition of compact length.

$$\tilde{\mathbf{v}}_r = \phi^{-1-\Phi} \mathbf{u} \quad , \quad \tilde{q}_f = \phi^{1/2} q_f \quad \text{and} \quad q = q_s + \phi(q_f - q_s)$$

while the pressure potentials are defined as

$$q_f = p_f - \rho_f g z \quad \text{and} \quad q_s = p_s - \rho_f g z$$

and we define reference physical variables as:

$$l_0 = \left(\frac{k_0 \mu_s}{\mu_f} \right)^{1/2} \quad \text{and} \quad q_0 = \rho_r g l_0 \quad \text{and} \quad w = \frac{k_0 \rho_r g}{\mu_f}$$

Hence we can reterive the two original physical pressure as:

$$p_f = q_0 \hat{q}_f + \rho_f g z$$

$$p_s = \frac{\hat{q} - \phi \hat{q}_f}{1 - \phi} q_0 + \rho_f g z$$

where $\hat{q}$ and $\hat{q}_f$ are dimensionless variables computed in the linear system.

Scale	Definition	Typical value	Units
V_0		3.2	cm y^{-1}
l_0	$\left(\frac{k_0 \mu_s}{\mu_f} \right)^{1/2}$		

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